A First Course in Causal Inference Ch.4–5

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Ch.4 Neymanian Repeated Sampling Inference in CRE

Ch.4 Neymanian Repeated Sampling Inference in CRE

- Neyman (1923) has proposed an unbiased point estimator and a conservative confidence interval based on the sampling distribution of the point estimator
- This chapter covers Neyman (1923)'s fundamental parts.

4.1 Finite population quantities

- Setup: CRE with n units (treatement group: n_1 units, control group: n_0 units); For $i = 1, ..., n$, potential outcomes $Y_i(1)$ and $Y_i(0)$ and individual effect $\tau_i = Y_i(1) - Y_i(0)$.
- We then obtain finite population means:

$$\overline{Y}(1) = \frac{1}{n}\sum_{i=1}^{n} Y_i(1), \quad \overline{Y}(0) = \frac{1}{n}\sum_{i=1}^{n} Y_i(0),$$

- variances:

$$S^2(1) = \frac{1}{n-1}\sum_{i=1}^{n}\{Y_i(1) - \overline{Y}(1)\}^2, \quad S^2(0) = \frac{1}{n-1}\sum_{i=1}^{n}\{Y_i(0) - \overline{Y}(0)\}^2,$$

- and covariance:

$$S(1,0) = \frac{1}{n-1}\sum_{i=1}^{n}\{Y_i(1) - \overline{Y}(1)\}\{Y_i(0) - \overline{Y}(0)\}.$$

4.1 Finite population quantities

- The individual effects have mean

$$\tau = \frac{1}{n}\sum_{i=1}^{n}\tau_i = \overline{Y}(1) - \overline{Y}(0),$$

- and variance

$$S^2(\tau) = \frac{1}{n-1}\sum_{i=1}^{n}(\tau_i - \tau)^2$$

We then have

Lemma 4.1

$$2S(1,0) = S^2(1) + S^2(0) - S^2(\tau).$$

4.2 Neyman (1923)'s theorem

- We are interested in estimating the average causal effect τ based on the (observed) data $(Z_i, Y_i)_{i=1}^n$ from a CRE.
- Based on the observed data, the sample means:

$$\widehat{\bar{Y}}(1) = \frac{1}{n_1}\sum_{i=1}^n Z_i Y_i, \quad \widehat{\bar{Y}}(0) = \frac{1}{n_0}\sum_{i=1}^n (1 - Z_i) Y_i,$$

- and the sample variances:

$$\widehat{S}^2(1) = \frac{1}{n_1 - 1}\sum_{i=1}^n Z_i\{Y_i - \widehat{\bar{Y}}(1)\}^2, \quad \widehat{S}^2(0) = \frac{1}{n_0 - 1}\sum_{i=1}^n (1 - Z_i)\{Y_i - \widehat{\bar{Y}}(0)\}^2.$$

- Obviously, there are no sample covariance and $S^2(\tau)$ because **we cannot observe both potential outcomes for each unit**.

4.2 Neyman (1923)'s theorem

Theorem 4.1

1. the difference-in-means estimator $\hat{\tau} = \hat{\bar{Y}}(1) - \hat{\bar{Y}}(0)$ is unbiased for τ:
$$\mathbb{E}[\hat{\tau}] = \tau.$$

2. $\hat{\tau}$ has variance
$$\mathrm{var}(\hat{\tau}) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n} = \frac{n_0}{n_1 n} S^2(1) + \frac{n_1}{n_0 n} S^2(0) + \frac{2}{n} S(1,0).$$

3. the variance estimator $\hat{V} = \frac{\hat{S}^2(1)}{n_1} + \frac{\hat{S}^2(0)}{n_0}$ is conservative for estimating $\mathrm{var}(\hat{\tau})$ in the sense that $\mathbb{E}[\hat{V}] - \mathrm{var}(\hat{\tau} = \frac{S^2(\tau)}{n}) \geq 0$ with equality holding if and only if $\tau_i = \tau$ for all units.

4.2 Neyman (1923)'s theorem

- The FRT works for any test statistic but Neyman (1923)'s theorem is only about the difference in means.
- If $S(1,0) > 0$, then those treated units also have relatively large control potential outcomes[1]. $\rightsquigarrow$ the observed outcomes under control are relatively small $\rightsquigarrow$ large $\widehat{\tau}$.
- If $S(1,0) < 0$, the those treated units have relatively small control potential outcomes[2]. $\rightsquigarrow$ the observed outcomes under control are relatively large $\rightsquigarrow$ small $\widehat{\tau}$.

[1] In other words, large control potential outcomes are not "observed."
[2] In other words, small control potential outcomes are not "observed."

4.2 Neyman (1923)'s theorem

Thoerem 4.2

Let $n \to \infty$ and $n_i \to \infty$. If $\dfrac{n_1}{n}$ has a limiting value in $(0,1)$, $\{S^2(1), S^2(0), S(1,0)\}$ have limiting values, and

$$\max_{1 \le i \le n} \frac{\{Y_i(1) - \overline{Y}(1)\}^2}{n} \to 0, \quad \max_{1 \le i \le n} \frac{\{Y_i(1) - \overline{Y}(0)\}^2}{n} \to 0, \tag{1}$$

then

$$\frac{\widehat{\tau} - \tau}{\sqrt{\mathrm{var}(\widehat{\tau})}} \xrightarrow[d]{} \mathcal{N}(0,1), \tag{2}$$

and

$$\widehat{S}^2(1) \xrightarrow[p]{} S^2(1), \quad \widehat{S}^2(1) \xrightarrow[p]{} S^2(0) \tag{3}$$

4.2 Neyman (1923)'s theorem

The intuitive understanding for Thm. 4.2 is:

- Eq. (1): potential outcomes for each unit are not extremely large to dominate the variability[3].
- Suppose a case one unit has an extremely large outcome, then $\widehat{\tau}$ depends on which group the unit belongs to: Eq. (2) does not hold.
- Eq. (3) ensures Neyman (1923)'s conservative CI for τ, because $\widehat{V} \geq \mathrm{var}(\widehat{\tau})$.

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[3] Formally, $\max_{1 \leq i \leq n} \| Y_i(1) - \overline{Y}(1) \| = o(\sqrt{n})$.

4.3 Proofs

Unbiasedness of $\widehat{\tau}$:

$$\widehat{\tau} = \frac{1}{n_1}\sum_{i=1}^{n} Z_i Y_i - \frac{1}{n_0}\sum_{i=1}^{n}(1-Z_i)Y_i = \frac{1}{n_1}\sum_{i=1}^{n} Z_i Y_i(1) - \frac{1}{n_0}\sum_{i=1}^{n}(1-Z_i)Y_i(0),$$

then

$$\begin{aligned}
\mathbb{E}[\widehat{\tau}] &= \mathbb{E}\left[\frac{1}{n_1}\sum_{i=1}^{n} Z_i Y_i(1) - \frac{1}{n_0}\sum_{i=1}^{n}(1-Z_i)Y_i(0)\right] \\
&= \frac{1}{n_1}\sum_{i=1}^{n} \mathbb{E}[Z_i]\, Y_i(1) - \frac{1}{n_0}\sum_{i=1}^{n}\mathbb{E}[1-Z_i]\, Y_i(0) \\
&= \frac{1}{n_1}\sum_{i=1}^{n}\frac{n_1}{n} Y_i(1) - \frac{1}{n_0}\sum_{i=1}^{n}\frac{n_0}{n} Y_i(0) = \frac{1}{n}\sum_{i=1}^{n} Y_i(1) - \frac{1}{n}\sum_{i=1}^{n} Y_i(0) = \tau. \quad \square
\end{aligned}$$

4.3 Proofs

$\operatorname{var}(\widehat{\tau})$: We write $\widehat{\tau}$ as

$$\widehat{\tau} = \sum_{i=1}^{n} Z_i \left\{ \frac{Y_i(1)}{n_1} + \frac{Y_i(0)}{n_0} \right\} - \frac{1}{n_0} \sum_{i=1}^{n} Y_i(0).$$

Then, the variance of $\widehat{\tau}$ follows:

$$\begin{aligned}
\operatorname{var}(\widehat{\tau}) &= \frac{n_1 n_0}{n(n-1)} \sum_{i=1}^{n} \left\{ \frac{Y_i(1)}{n_1} + \frac{Y_i(0)}{n_0} - \frac{\overline{Y}(1)}{n_1} - \frac{\overline{Y}(0)}{n_0} \right\}^2 \\
&= \frac{n_1 n_0}{n(n-1)} \left[\frac{1}{n_1^2} \sum_{i=1}^{n} \{Y_i(1) - \overline{Y}(1)\}^2 + \frac{1}{n_0^2} \sum_{i=1}^{n} \{Y_i(0) - \overline{Y}(0)\}^2 \right. \\
&\qquad \left. + \frac{2}{n_1 n_0} \sum_{i=1}^{n} \{Y_i(1) - \overline{Y}(1)\}\{Y_i(0) - \overline{Y}(0)\} \right] \\
&= \frac{n_0}{n_1 n} S^2(1) + \frac{n_1}{n_0 n} S^2(0) + \frac{2}{n} S(1,0).
\end{aligned}$$

4.3 Proofs

Here, the first equality holds because of the following Lemma:

Lemma C.2

Under simple random sampling, the sample means are unbiased for the population means:

$$\mathbb{E}\left[\widehat{\bar{c}}\right] = \bar{c}, \quad \mathbb{E}\left[\widehat{\bar{d}}\right] = \bar{d}.$$

Their variances and covariance are

$$\operatorname{var}\left(\widehat{\bar{c}}\right) = \frac{n_0}{nn_1} S_c^2, \quad \operatorname{var}\left(\widehat{\bar{d}}\right) = \frac{n_0}{nn_1} S_d^2, \quad \operatorname{cov}\left(\widehat{\bar{c}}, \widehat{\bar{d}}\right) = \frac{n_0}{nn_1} S_c d.$$

Let $A_i = \dfrac{Y_i(1)}{n_1} + \dfrac{Y_i(0)}{n_0}$, then $\mathrm{var}(\widehat{\tau}) = \mathrm{var}\left(\sum_{i=1}^n Z_i A_i\right)$. In a CRE, we draw a simple random sample of size n_1 for a treatment group. Then, $\overline{A} = \dfrac{1}{n_1}\sum_{i=1}^n Z_i A_i$. Following Lemma C.2, $\mathrm{var}(\overline{A}) = \dfrac{n_0}{nn_1} S_A^2$. Hence,

$$\mathrm{var}(\widehat{\tau}) = \mathrm{var}\left(\sum_{i=1}^n Z_i A_i\right) = \mathrm{var}(n_1 \overline{A}) = n_1^2 \, \overline{\mathrm{A}} = n_1^2 \frac{n_0}{nn_1} S_A^2 = \frac{n_0 n_1}{n} S_A^2.$$

Because $S_A^2 = \dfrac{1}{n-1}\sum_{i=1}^n \{A_i - \overline{A}\}^2$, we obtain

$$\mathrm{var}(\widehat{\tau}) = \frac{n_0 n_1}{n(n-1)} \sum_{i=1}^n \left\{ \frac{Y_i(1)}{n_1} + \frac{Y_i(0)}{n_0} - \frac{\overline{Y}(1)}{n_1} - \frac{\overline{Y}(0)}{n_0} \right\}^2$$

From Lemma 4.1, we also write the variance as

$$\mathrm{var}(\widehat{\tau}) = \frac{n_0}{n_1 n} S^2(1) + \frac{n_1}{n_0 n} S^2(0) + \frac{1}{n}(S^2(1) + S^2(0) - S^2(\tau)) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}. \quad \square$$

4.4 Regression analysis of the CRE

- Practitioners often use regression-based inference for the average causal effect τ.
- A standard approach is OLS:

$$(\widehat{\alpha}, \widehat{\beta}) = \arg\min_{(a,b)} \sum_{i=1}^{n} (Y_i - a - bZ_i)^2.$$

- Here, $\widehat{\tau} = \widehat{\beta}$.
- However, the usual variance estimator from the OLS equals

$$\widehat{V}_{\text{OLS}} = \frac{n(n_1 - 1)}{(n-2)n_1 n_0} \widehat{S}^2(1) + \frac{n(n_0 - 1)}{(n-2)n_1 n_0} \widehat{S}^2(0) \approx \frac{\widehat{S}^2(1)}{n_0} + \frac{\widehat{S}^2(0)}{n_1},$$

where the approximation holds with large n_1 and n_0 (it can differ).
- EHW robust variance estimator is closer to $\widehat{V}$ and HC2 variianct of the EHW robust variance estimator is identical to $\widehat{V}$.

Ch.5 Stratification and Post-Stratification in Randomized Experiments

Ch.5 Stratification and Post-Stratification in Randomized Experiments

This chapter explains the meaning of the following quote:

Block what you can and randomize what you cannot.
– Box et al. (1978)

5.1 Stratification

- A CRE may generate an undesired treatment allocation by chane.
- Setup: a CRE wuth a discrete covariate $X_i \in \{1, \dots, K\}$; define $n_{[k]} = \#\{i : X_i = k\}$ and $\pi_{[k]} = n_{[k]}/n$ as the number and proportion of units in stratum k ($k = 1, \dots, K$).
- Then, $n_{[k]1} = \#\{i : X_i = k, Z_i = 1\}$ and $n_{[k]0} = \#\{i : X_i = k, Z_i = 0\}$.
- With positive probability, $n_{[k]1}$ or $n_{[k]0}$ is zero for some k: **it is possible that some strata only have treated or control units.**
- So, the proportions of units in stratum k are different across the treatment and control groups although on average their difference is zero:

$$\mathbb{E}\left[\frac{n_{[k]1}}{n_1} - \frac{n_{[k]0}}{n_0}\right] = 0. \tag{4}$$

Proof for Eq. (4).
We also write $n_{[k]1}$ and $n_{[k]0}$ as $n_{[k]1} = \sum_{i:X_i=k} Z_i$ and $n_{[k]0} = \sum_{i:X_i=k}(1 - Z_i)$, respectively. We then obtain

$$\mathbb{E}\left[\frac{n_{[k]1}}{n_1}\right] = \frac{1}{n_1}\mathbb{E}[n_{[k]1}] = \frac{1}{n_1}\mathbb{E}\left[\sum_{i:X_i=k} Z_i\right] = \frac{1}{n_1}\sum_{i:X_i=k}\mathbb{E}[Z_i] = \frac{1}{n_1} n_{[k]}\frac{n_1}{n} = \frac{n_{[k]}}{n}.$$

Similarly,

$$\mathbb{E}\left[\frac{n_{[k]0}}{n_0}\right] = \frac{n_{[k]}}{n}.$$

Hence, $\mathbb{E}\left[\frac{n_{[k]1}}{n_1} - \frac{n_{[k]0}}{n_0}\right] = 0.$ $\square$

5.1 Stratification

- When $\dfrac{n_{[k]1}}{n_1} - \dfrac{n_{[k]0}}{n_0}$ is large for some strata k's, the threatment and control groups have undesirable covariate imbalance.
- In an extreme example, suppose $k = 2$ and $n = 8$, $n_1 = 4$, $n_0 = 4$, $n_{[1]} = 4$, and $n_{[2]} = 4$. There exists a case $n_{[1]1} = 4$, $n_{[1]0} = 0$, $n_{[2]1} = 0$, and $n_{[2]0} = 4$. Here there is undesirebale covariate imbalance amongst both groups.
- To avoid such covariate imbalance in experiments, we introduce *stratified randomized experiments* (SRE).

5.1 Stratification

Definition 5.1 (SRE)

Fix the $n_{[k]1}$'s or $n_{[k]0}$'s. We conduct K independent CREs within the K strata of a discrete covariate X.

- The total number of randamizations in an SRE equals

$$\prod_{k=1}^{K} \binom{n_{[k]}}{n_{[k]1}},$$

  and each feasible randomization has equal probability.
- Within stratum k, the proportion of units receiving the treatment is

$$e_{[k]} = \frac{n_{[k]1}}{n_{[k]}},$$

  which is also called the *propensity score*.

5.1 Stratification

An SRE is different from a CRE:

1. $$\prod_{k=1}^{K} \binom{n_{[k]}}{n_{[k]1}} < \binom{n}{n_1}.$$

   - In the previous example, in a CRE the number of allocation is $\binom{8}{4} = 70$ while that of allocation is $\binom{4}{2}\binom{4}{2} = 36$. Hence $70 > 36$.

2. $e_{[k]}$ is fixed in an SRE but random in a CRE
   - It says the excatly same thing to the previous illustration.

5.1 Stratification

- For stratum k, we have the stratum-specific average causal effect

$$\tau_{[k]} = \frac{1}{n_{[k]}} \sum_{X_i=k} \tau_i.$$

- The average causal effect is[4]

$$\tau = \frac{1}{n}\sum_{i=1}^{n}\tau_i = \frac{1}{n}\sum_{k=1}^{K}\sum_{X_i=k}\tau_i = \frac{1}{n}\sum_{k=1}^{K} n_{[k]}\tau_{[k]} = \sum_{k=1}^{K}\frac{n_{[k]}}{n}\tau_{[k]} = \sum_{k=1}^{K}\pi_{[k]}\tau_{[k]}.$$

[4] The third equality follows from the equation above: $\sum_{X_i=k}\tau_i = n_{[k]}\tau_{[k]}$.

5.2 FRT

- The FRT in an SRE.
- The sharp null hypothesis: $H_{0F}: Y_i(1) = Y_i(0)$ for all units $i = 1, \dots, n$.
- Following the Def. 5.1, when we simulate the treatment vector, we must permute the treatment indicators within strata of X.
- We should choose test statistics that can reflect the nature of the SRE.

5.2 FRT

Example 5.1 Stratified estimator.
Motivated by estimating τ, we can use the following stratified estimator in the FRT:

$$\widehat{\tau}_S = \sum_{k=1}^{K} \pi_{[k]} \widehat{\tau}_{[k]},$$

where

$$\widehat{\tau}_{[k]} = \frac{1}{n_{[k]1}} \sum_{i=1}^{n} \mathbb{I}(X_i = k, Z_i = 1) Y_i - \frac{1}{n_{[k]0}} \sum_{i=1}^{n} \mathbb{I}(X_i = k, Z_i = 0) Y_i$$

is the staratum-specific difference-in-means within stratum k.

Example 5.2 Studentized stratified estimator.
Motivated by the studentized statistic in the simple two-sample problem, we can use the following studentized statistic for the stratified estimator in the FRT:

$$t_S = \frac{\hat{\tau}_S}{\sqrt{\hat{V}_S}},$$

with

$$\hat{V}_S = \sum_{k=1}^{K} \pi_{[k]}^2 \left(\frac{\hat{S}_{[k]}^2(1)}{n_{[k]1}} + \frac{\hat{S}_{[k]}^2(0)}{n_{[k]0}} \right)$$

where $\hat{S}_{[k]}^2(1)$ and $\hat{S}_{[k]}^2(0)$ are the stratum-specific sample variances of the outcomes under the treatment and control, respectively.

Example 5.3 Combining Wilcoxon rank-sum statistics.
We first compute the Wilcoxon rank sum statistic $W_{[k]}$[5] within stratum k and then combine them as

$$W_S = \sum_{k=1}^{K} c_{[k]} W_{[k]}.$$

For $c_{[k]}$, there are two proposed methods:

$$c_{[k]} = \frac{1}{n_{[k]1} n_{[k]0}},$$
$$c_{[k]} = \frac{1}{n_{[k]} + 1}.$$

[5] Recall the Wilcoxon rank sum statistic is defined as $W = \sum_{i=1}^{n} Z_i R_i$, where $R_i = \#\{j : Y_j \leq Y_i\}$.

Example 5.4 Hodges Jr and Lehmann (2011)'s aligned rank statistics.
Hodges Jr and Lehmann (2011) proposed a test statistic that make more comparisons across strata after standardizing the outcomes. First,

$$\widetilde{Y}_i = Y_i - \overline{Y}_{[k]}$$

with $\overline{Y}_{[k]} = \frac{1}{n_{[k]}} \sum_{X_i=k} Y_i$, then obtaining the ranks $(\widetilde{R}_1, \ldots, \widetilde{R}_n)$ of the pooled outcomes $(\widetilde{Y}_1, \ldots, \widetilde{Y}_n)$, and finally constructing the test statistic

$$\widetilde{W} = \sum_{i=1}^{n} Z_i \widetilde{R}_i.$$

5.3 Neymanian inference

- Statistical inference for an SRE builds on the fact that it essentially consists of K independent CREs.
- Within sratum k, the difference-in-means $\widehat{\tau}_{[k]}$ is unbiased for $\tau_{[k]}$ with variances

$$\operatorname{var}(\widehat{\tau}_{[k]}) = \frac{S^2_{[k]}(1)}{n_{[k]1}} + \frac{S^2_{[k]}(0)}{n_{[k]0}} - \frac{S^2_{[k]}(\tau)}{n_{[k]}}.$$

- Hence, the stratified estimator $\widehat{\tau}_S = \sum_{k=1}^K \pi_{[k]}\widehat{\tau}_{[k]}$ is unbiased for $\tau = \sum_{k=1}^K \pi_{[k]}\tau_{[k]}$ with variance

$$\operatorname{var}(\widehat{\tau}_S) = \sum_{k=1}^K \pi^2_{[k]} \operatorname{var}(\widehat{\tau}_{[k]}).$$

5.3 Neymanian inference

- Similarly, we obtain a conservative variance estimator

$$\widehat{V}_S = \sum_{k=1}^{K} \pi_{[k]}^2 \left(\frac{\widehat{S}^2_{[k]}(1)}{n_{[k]1}} + \frac{\widehat{S}^2_{[k]}(0)}{n_{[k]0}} \right),$$

  with sample variances of the outcomes $\widehat{S}^2_{[k]}(1)$ and $\widehat{S}^2_{[k]}(0)$.
- We can also construct a Wald-type $1-\alpha$ CI for τ and p-values.
- In short, **SRE is essentially K independent CREs**.

5.3 Neymanian inference

- What are the benefits of the SRE compared with the CRE?
- Benefits from the covariate balance in the SRE.
- Better covariate balance results in better estimation precision of the average causal effect.

5.3 Neymanian inference

Assume that $e_{[k]} = e$ for all k. Then

$$\widehat{\tau} = \widehat{\tau}_S. \qquad (5)$$

▸ Proof

5.3 Neymanian inference

To compare $\mathrm{var}_{CRE}(\widehat{\tau})$ and $\mathrm{var}_{SRE}(\widehat{\tau}_S)$, we first derive $S^2(1)$, $S^2(0)$ and $S^2(\tau)$. We then obtain

$$S^2(1) = \sum_{k=1}^{K} \left[\frac{n_{[k]} - 1}{n - 1} S^2_{[k]}(1) + \frac{n_{[k]}}{n - 1} \{ \overline{Y}_{[k]}(1) - \overline{Y}(1) \}^2 \right],$$

$$S^2(0) = \sum_{k=1}^{K} \left[\frac{n_{[k]} - 1}{n - 1} S^2_{[k]}(0) + \frac{n_{[k]}}{n - 1} \{ \overline{Y}_{[k]}(0) - \overline{Y}(0) \}^2 \right],$$

$$S^2(\tau) = \sum_{k=1}^{K} \left[\frac{n_{[k]} - 1}{n - 1} S^2_{[k]}(\tau) + \frac{n_{[k]}}{n - 1} \{ \tau_{[k]} - \tau \}^2 \right].$$

▸ Proof

Hence,

$$
\begin{aligned}
&\operatorname{var}_{CRE}(\widehat{\tau}) \\
&= \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n} \\
&= \sum_{k=1}^{K} \left[\frac{n_{[k]} - 1}{(n-1)n_1} S^2_{[k]}(1) + \frac{n_{[k]} - 1}{(n-1)n_0} S^2_{[k]}(0) - \frac{n_{[k]} - 1}{(n-1)n} S^2_{[k]}(\tau) \right] \\
&+ \sum_{k=1}^{K} \left[\frac{n_{[k]}}{(n-1)n_1} \{\overline{Y}_{[k]}(1) - \overline{Y}(1)\}^2 + \frac{n_{[k]}}{(n-1)n_0} \{\overline{Y}_{[k]}(0) - \overline{Y}(0)\}^2 - \frac{n_{[k]}}{(n-1)n} \{\tau_{[k]} - \tau\}^2 \right]
\end{aligned}
$$

5.3 Neymanian inference

With large $n_{[k]}$'s, $n_{[k]} - 1 \approx n_{[k]}$ and $n - 1 \approx n$ then

$$\begin{aligned}
\text{var}_{CRE}(\widehat{\tau}) \approx & \sum_{k=1}^{K} \left[\frac{\pi_{[k]}}{n_1} S^2_{[k]}(1) + \frac{\pi_{[k]}}{n_0} S^2_{[k]}(0) - \frac{\pi_{[k]}}{n} S^2_{[k]}(\tau) \right] \\
& + \sum_{k=1}^{K} \left[\frac{\pi_{[k]}}{n_1} \{\overline{Y}_{[k]}(1) - \overline{Y}(1)\}^2 + \frac{\pi_{[k]}}{n_0} \{\overline{Y}_{[k]}(0) - \overline{Y}(0)\}^2 - \frac{\pi_{[k]}}{n} \{\tau_{[k]} - \tau\}^2 \right].
\end{aligned}$$

Following the assumption $e_{[k]} = e$, we write $\text{var}_{SRE}(\widehat{\tau})$ as

$$\begin{aligned}
\text{var}_{SRE}(\widehat{\tau}_S) &= \sum_{k=1}^{K} \pi^2_{[k]} \left[\frac{S^2_{[k]}(1)}{n_{[k]1}} + \frac{S^2_{[k]}(0)}{n_{[k]0}} - \frac{S^2_{[k]}(\tau)}{n_{[k]}} \right] \\
&= \sum_{k=1}^{K} \left[\frac{\pi_{[k]}}{n_1} S^2_{[k]}(1) + \frac{\pi_{[k]}}{n_0} S^2_{[k]}(0) - \frac{\pi_{[k]}}{n} S^2_{[k]}(\tau) \right].
\end{aligned}$$

With large $n_{[k]}$'s, approximately,

$$
\begin{aligned}
&\text{var}_{CRE}(\widehat{\tau}) - \text{var}_{SRE}(\widehat{\tau}_S) \\
&\approx \sum_{k=1}^{K} \left[\frac{\pi_{[k]}}{n_1} \{\overline{Y}_{[k]}(1) - \overline{Y}(1)\}^2 + \frac{\pi_{[k]}}{n_0} \{\overline{Y}_{[k]}(0) - \overline{Y}(0)\}^2 - \frac{\pi_{[k]}}{n} \{\tau_{[k]} - \tau\}^2 \right] \\
&\geq 0.
\end{aligned} \tag{6}
$$

The equality holds only in a case that for $k = 1, \ldots, K$,

$$
\sqrt{\frac{n_0}{n_1}} \{\overline{Y}_{[k]}(1) - \overline{Y}(1)\}^2 + \sqrt{\frac{n_1}{n_0}} \{\overline{Y}_{[k]}(0) - \overline{Y}(0)\} = 0.
$$

Proof

5.3 Neymanian inference

This result has several implications:

- An SRE is generally more efficient than a CRE, especially when the stratifying covariate is predictive of the potential outcomes.
- In the extreme case where the stratifying covariate contains little information about the potential outcomes (that is, the potential outcomes are nearly the same across strata), an SRE provides little or no efficiency gain over a CRE.
- This comparison relies on the large-strata approximation.
- Therefore, the result does not necessarily hold in finite samples; in some cases, an SRE can even be less efficient than a CRE.

5.4 Post-stratification in a CRE

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Proof of Eq. (5)

Recall

$$\widehat{\tau} = \frac{1}{n_1}\sum_{i=1}^{n} Z_i Y_i - \frac{1}{n_0}\sum_{i=1}^{n}(1-Z_i)Y_i, \quad \widehat{\tau}_S = \sum_{k=1}^{K}\pi_{[k]}\widehat{\tau}_{[k]},$$

where

$$\widehat{\tau}_{[k]} = \frac{1}{n_{[k]1}}\sum_{i:X_i=k} Z_i Y_i - \frac{1}{n_{[k]0}}\sum_{i:X_i=k}(1-Z_i)Y_i, \quad \pi_{[k]} = \frac{n_{[k]}}{n}$$

Then,

$$\begin{aligned}\widehat{\tau}_S &= \sum_{k=1}^{K}\frac{1}{n\cdot\frac{n_{[k]1}}{n_{[k]}}}\sum_{i:X_i=k} Z_i Y_i - \sum_{k=1}^{K}\frac{1}{n\cdot\frac{n_{[k]0}}{n_{[k]}}}\sum_{i:X_i=k}(1-Z_i)Y_i \\ &= \frac{1}{n_1}\sum_{k=1}^{K}\sum_{i:X_i=k} Z_i Y_i - \frac{1}{n_0}\sum_{k=1}^{K}\sum_{i:X_i=k}(1-Z_i)Y_i = \widehat{\tau}. \quad \square\end{aligned}$$

The second equality follows from the assumption $e = e_{[k]} = n_{[k]1}/n_{[k]}$.

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Variances Derivation

$$
\begin{aligned}
S^2(1) &= \frac{1}{n-1}\sum_{i=1}^{n}\{Y_i(1)-\overline{Y}(1)\}^2 \\
&= \frac{1}{n-1}\sum_{k=1}^{K}\sum_{i:X_i=k}\{Y_i(1)-\overline{Y}_{[k]}(1)+\overline{Y}_{[k]}(1)-\overline{Y}(1)\}^2 \\
&= \frac{1}{n-1}\sum_{k=1}^{K}\sum_{i:X_i=k}[\{Y_i(1)-\overline{Y}_{[k]}(1)\}^2+\{\overline{Y}_{[k]}(1)-\overline{Y}(1)\}^2] \\
&= \sum_{k=1}^{K}\left[\frac{n_{[k]}-1}{n-1}S^2_{[k]}(1)+\frac{n_{[k]}}{n-1}\{\overline{Y}_{[k]}(1)-\overline{Y}(1)\}^2\right]. \quad \square
\end{aligned}
$$

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Proof of Eq. (6)

Let $A_k = \overline{Y}_{[k]}(1) - \overline{Y}()1$ and $B_k = \overline{Y}_{[k]}(0) - \overline{Y}(0)$. Because $\tau_{[k]} = \overline{Y}_{[k]}(1) - \overline{Y}_{[k]}(0)$ and $\tau = \overline{Y}(1) - \overline{Y}(0)$, we write the difference as:

$$\begin{aligned}
\sum_{k=1}^{K}\left[\frac{\pi_{[k]}}{n_1}A_k^2 + \frac{\pi_{[k]}}{n_0}B_k^2 - \frac{\pi_{[k]}}{n}(A_k - B_k)^2\right] &= \sum_{k=1}^{K}\frac{\pi_{[k]}}{n}\left[\frac{n}{n_1}A_k^2 + \frac{n}{n_0}B_k^2 - (A_k - B_k)^2\right] \\
&= \sum_{k=1}^{K}\frac{\pi_{[k]}}{n}\left[\left(\frac{n}{n_1} - 1\right)A_k^2 + \left(\frac{n}{n_0} - 1\right)B_k^2 + 2A_kB_k\right] \\
&= \sum_{k=1}^{K}\frac{\pi_{[k]}}{n}\left[\frac{n_0}{n_1}A_k^2 + \frac{n_1}{n_0}B_k^2 + 2A_kB_k\right] \\
&= \sum_{k=1}^{K}\frac{\pi_{[k]}}{n}\left\{\sqrt{\frac{n_0}{n_1}}A_k + \sqrt{\frac{n_1}{n_0}}B_k\right\}^2 \geq 0. \quad \square
\end{aligned}$$

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